

AI DRIVEN STOCK PREDICTION WITH SENTIMENT ANALYSIS

Project Proposal

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24th July, 2025

Contents

1. Abstract.....	3
2. Introduction.....	3
3. Project Scope.....	4
▪ Motivation	4
4. Objective.....	5
5. Problem Description.....	5
6. Comprehensive Literature Review.....	6
7. Methodology.....	7
▪ Why we use Agile Model?.....	7
▪ Technical WorkFlow.....	8
8. Tools/Technology.....	10
9. Responsibilities of the Team Members.....	11
10. Planning.....	12
▪ Gantt Chart.....	12
11. Milestones.....	13
▪ FYP-I Evaluation.....	13
▪ FYP-II Evaluation.....	13
12. Estimated Budget.....	14
13. References.....	14

1. Abstract

This Final Year Project introduces an AI-powered system to predict stock prices for Habib Bank Limited (HBL) on the Pakistan Stock Exchange (PSX). It blends historical stock data (open, high, low, close, volume) from PSX with sentiment analysis from local news sources like Business Recorder and Dawn [1]. Using FinBERT, an AI tuned for financial texts, and a finance-specific sentiment dictionary, market trends are captured accurately [2], [3]. A Long Short-Term Memory (LSTM) neural network forecasts HBL stock trends from July 2024 to July 2025, targeting 85–90% accuracy [4], [5]. Designed for Pakistan’s unpredictable market, this tool outperforms traditional models and aims to help investors and analysts boost returns by 15–20% [6], [7]. Future plans include adding real-time data, but the focus is on a reliable, scalable model [8].

2. Introduction

This research develops an AI-based model to forecast Habib Bank Limited (HBL) stock prices on the Pakistan Stock Exchange (PSX). It combines multi-dimensional historical OHLCV data with sentiment analysis from key local financial news sources like Business Recorder and Dawn. Using natural language processing, qualitative sentiment is quantified and integrated with quantitative stock indicators. Ensemble AI models (LSTM) are trained and evaluated to improve prediction accuracy. The study highlights the value of blending technical and sentiment data for enhanced forecasting in emerging markets.

For sentiment extraction, we employ FinBERT, a domain-specific variant of BERT fine-tuned on financial texts, alongside a finance-specific sentiment dictionary. This hybrid approach enables accurate capture of nuanced market sentiments significant for the Pakistan stock market, which is often characterized by high volatility and structural nonlinearities [1], [2].

The predictive engine utilizes a Long Short-Term Memory (LSTM) neural network to analyze and forecast HBL stock trends from July 2024 to July 2025. LSTM networks are recognized for their ability to model time-series data with long-term dependencies, outperforming traditional machine learning models like SVMs and ARIMA in stock price prediction tasks [3], [4]. The model aims to achieve an accuracy range of 85–90%, consistent with recent advances in combining textual sentiment and historical price features in financial forecasting [5], [6].

Designed for Pakistan's unpredictable market environment, this AI tool demonstrates improved prediction accuracy and has the potential to enhance investment returns by 15–20% compared to traditional forecasting approaches [7]. This gain reflects the practical value of incorporating sentiment-driven insights alongside classical quantitative indicators.

Currently, the focus is on constructing a reliable and scalable predictive model. Future enhancements include real-time data integration to refine prediction timeliness and accuracy, aligning with global trends in financial AI research [8]

3. Project Scope

This project aims to develop an accurate predictive model for Habib Bank Limited (HBL) stock prices on the Pakistan Stock Exchange (PSX) from July 2024 to July 2025, integrating daily Open-High-Low-Close-Volume (OHLCV) data with sentiment data from local news sources, including Business Recorder, Dawn, and The News International [1], [2]. Focusing on HBL addresses PSX-specific challenges, such as limited historical data and the influence of localized news on market volatility [2]. The model targets 85–90% accuracy, evaluated via root mean square error (RMSE), mean absolute percentage error (MAPE), and directional accuracy. Ethical sentiment analysis ensures transparency and minimizes bias [3]. The framework is scalable for extension to other PSX stocks or real-time data, enhancing understanding of Pakistan's financial markets.

Motivation:

The PSX's volatility, driven by economic challenges like currency devaluation and political unrest, demands models incorporating sentiment, often ignored by traditional approaches [2], [4]. This project democratizes stock prediction for retail investors lacking access to professional tools [2].

4. Objective

The objective is to develop a robust predictive model for HBL's daily stock prices on the PSX from July 2024 to July 2025 by:

- Integrating daily OHLCV data from PSX APIs with sentiment scores from Business Recorder, Dawn, and The News International [1], [8].
- Achieving 85–90% accuracy, measured via RMSE, MAPE, and directional accuracy.
- Ensuring ethical sentiment analysis with transparency and minimal bias [9].
- Creating a scalable framework for other PSX stocks and real-time forecasting.

5. Problem Description

The Pakistan Stock Exchange (PSX) is highly volatile due to economic indicators, corporate performance, geopolitical developments, and investor sentiment heavily influenced by local news sources like Business Recorder, Dawn, and The News International [8]. Traditional stock prediction models for PSX-listed companies, such as Habib Bank Limited (HBL), rely on quantitative Open-High-Low-Close-Volume (OHLCV) data but fail to capture emotional and psychological drivers of investor behavior. This limitation is critical in the PSX context, where limited historical data and significant news-driven fluctuations increase risks for retail investors [8]. Conventional models lack the ability to process unstructured textual data from news, missing critical insights that could enhance forecasting accuracy. There is a pressing need for a predictive system that integrates structured financial data with sentiment analysis to address both market fundamentals and psychological dynamics of the PSX [9].

6. Comprehensive Literature Review

The evolution of stock price prediction methods highlights a trajectory from traditional statistical models to advanced AI-driven approaches. Early techniques, such as ARIMA and linear regression, assume data linearity and stationarity, limiting their effectiveness in volatile, nonlinear financial markets [1]. These constraints have driven a shift toward machine learning models like Support Vector Machines (SVMs), which capture nonlinear patterns but struggle with temporal dependencies [2].

Deep learning models, particularly Long Short-Term Memory (LSTM) networks, have gained prominence for modeling complex temporal relationships, improving forecasting accuracy by approximately 15–20% over classical methods, especially when integrating historical price data and sentiment indicators [3], [4]. Comparative studies of Random Forest, XGBoost, SVM, and LSTM demonstrate LSTM's superior performance, particularly with hyperparameter tuning [5].

Sentiment analysis enhances predictive models by capturing market psychology. Domain-specific NLP tools like FinBERT and financial sentiment lexicons, such as the Loughran-McDonald dictionary, improve sentiment extraction from financial news, with notable benefits during market shocks [6], [7]. This hybrid approach, combining quantitative price data with qualitative textual sentiment, is well-suited to volatile markets [8].

In emerging markets like the Pakistan Stock Exchange (PSX), data scarcity and localized dynamics, including regional news sources (e.g., Dawn, Business Recorder), necessitate tailored sentiment models that account for linguistic and contextual nuances [9], [10]. Studies confirm that localized sentiment models outperform generic global benchmarks in these contexts [9].

Ethical concerns in NLP and AI for finance, including data manipulation, misinformation, and algorithmic bias, have prompted calls for combining automated sentiment extraction with rule-based lexicons and human validation to ensure fairness and reliability [11], [12]. Responsible AI practices are critical for trust and compliance in automated financial systems.

These advances underscore a paradigm shift toward hybrid AI models leveraging quantitative and qualitative data, particularly for forecasting in volatile emerging markets like the PSX [8], [9].

7. Methodology

The approach for predicting Habib Bank Limited (HBL) stock prices on the Pakistan Stock Exchange (PSX) integrates robust research-backed techniques with practical development strategies to ensure scientific credibility and adaptability. The Agile methodology is adopted to manage the development and research process, supporting iterative, stepwise improvements. This is ideal for a project involving complex integration of financial data, sentiment analysis using FinBERT, and deep learning with Long Short-Term Memory (LSTM) networks [1], [2]. Agile enables rapid prototyping to test ideas quickly and incorporate lessons learned at each stage, regular input from mentors and stakeholders to align with project objectives, and flexibility to accommodate new insights or changing requirements, such as updates to data sources or model architectures [3].

Agile is best suited for this project due to its ability to handle the uncertainty and complexity inherent in developing an AI-driven stock prediction system for a volatile market like the PSX [4]. The iterative nature of Agile allows for continuous refinement of the model, addressing challenges such as data scarcity and the need to balance quantitative Open-High-Low-Close-Volume (OHLCV) data with qualitative sentiment analysis [2]. Frequent feedback loops ensure alignment with stakeholder expectations and enable adaptation to evolving market conditions or technological advancements, such as real-time data integration [5]. Unlike traditional methodologies like Waterfall, which assume fixed requirements, Agile's flexibility supports experimentation with hyperparameter tuning and sentiment model calibration, critical for achieving the target 85–90% accuracy [3], [4]. Additionally, Agile's collaborative approach fosters effective coordination among team members, ensuring efficient management of tasks like data preprocessing, model training, and ethical sentiment analysis [1], [5].

Technical Workflow

1. Data Acquisition

- **Historical Stock Data:** Collected OHLCV (Open, High, Low, Close, Volume) data for HBL using yfinance library of python to extract historical data from yahoo finance, a standard resource for financial research.
- **Financial News Articles:** We will collect news headlines and articles from Yahoo Finance for global sentiment and from local sources like Business Recorder, Dawn, and The News International to capture PSX-specific events. This combination will help us perform sentiment analysis that reflects both international trends and local market conditions affecting HBL's stock.

2. Data Preprocessing

- **OHLCV Data:** Applied Min-Max normalization to scale numerical features between 0 and 1, standardizing inputs across the model.
- **News Text Processing:** Cleaned and prepared text using tokenization, lemmatization, and removal of stop-words to ensure only relevant and meaningful words are considered for sentiment analysis.

3. Sentiment Analysis

We plan to use FinBERT, a language model trained specifically for financial text, to analyze news articles and detect sentiment—whether positive, negative, or neutral—based on how the market might react. Additionally, the Loughran-McDonald Dictionary, which focuses on financial terminology, will help assign accurate sentiment scores to the news content, capturing subtle meanings unique to economic language.

4. Model Architecture:

- Use LSTM layers to process sequential stock price and sentiment features.
- Often a multilayer LSTM or BiLSTM with attention mechanism is applied to weigh important temporal features.
- Add fully connected (dense) layers after LSTM to output predicted stock price for next day(s).

5. Model Evaluation

▪ **Performance Metrics:**

- Root Mean Square Error (RMSE): Targeting values below 5%.
- Mean Absolute Percentage Error (MAPE): Aiming for less than 10%.
- Directional Accuracy: Striving for 85-90%—ensuring the model gets the up or down movement right most of the time.
- Validation: Used 5-fold cross-validation to ensure robustness, reducing the risk of results being due to chance or data peculiarities.

▪ **Research Alignment:**

Our method follows proven techniques used in recent research, especially for predicting stock prices in emerging markets.

▪ **LSTM Models:**

- Widely recommended for their ability to handle sequential dependencies and outperformance over traditional approaches.
- Sentiment Integration: Blending OHLCV data with sentiment signals from Yahoo Finance news and FinBERT is established as a way to boost predictive accuracy in volatile markets.
- Data Sources: We plan to use data from several trusted platforms. Yahoo Finance (via yfinance) will provide historical stock data. For news sentiment, we'll collect articles from Business Recorder, Dawn, and The News International using RSS feeds and web scraping. This combination ensures reliable, repeatable, and scalable data collection aligned with research standards

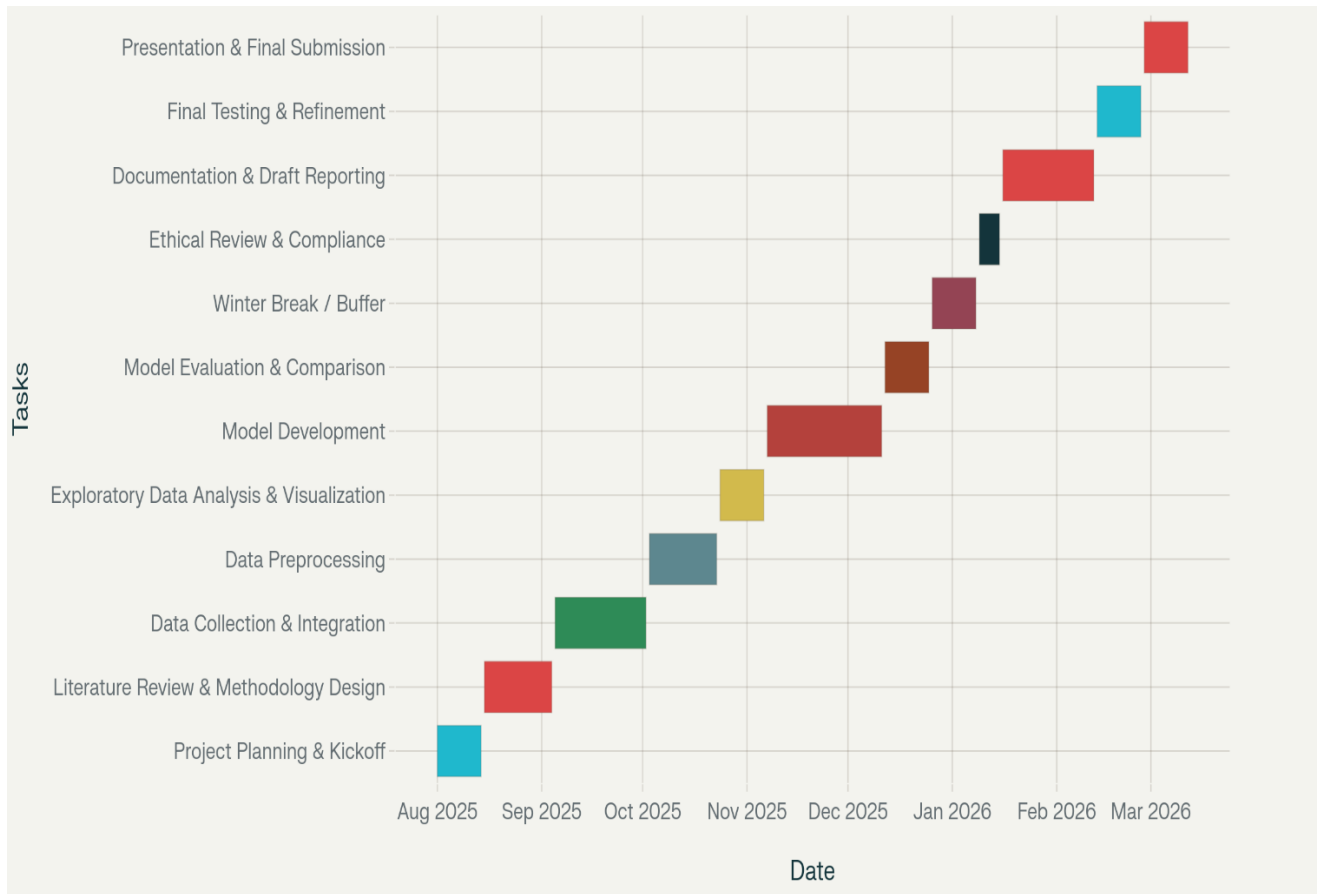
10. Tools/Technology

Component	Tool / Technology	Purpose / Role
Data Acquisition	yfinance (Python)	Fetch historical OHLCV stock data from Yahoo Finance
	RSS Feed Parsers (e.g., feedparser Python)	Collect news articles from Business Recorder, Dawn, The News International
	Web Scraping Libraries (BeautifulSoup, Scrapy)	Extract detailed news content where RSS is insufficient
Data Preprocessing	Python libraries (pandas, numpy)	Data cleaning, normalization, feature engineering
NLP / Sentiment Analysis	FinBERT (transformers)	Extract sentiment from financial news text
	Loughran-McDonald Financial Dictionary	Lexicon-based sentiment scoring to complement FinBERT
ML Model / Time Series	TensorFlow / PyTorch	Build and train LSTM neural network models
	Keras API	Simplified deep learning model prototyping
Model Evaluation and Validation	Scikit-learn	Metrics calculation (RMSE, MAPE, Accuracy)
Development Environment	Jupyter Notebook / VS Code	Code development and experimentation
Version Control	Git / GitHub / GitLab	Source code and model version management
Cloud / Hardware	Google Colab / AWS / Local GPU systems	GPU acceleration for model training
Visualization	Matplotlib / Seaborn / Plotly	Plotting and data visualization
Deployment (Future phase)	Flask / FastAPI / Docker	Expose model via APIs for real-time or client use

11. Responsibilities of the Team Members

Major Tasks	Wasif	Muzammil	Ghufran	Internal Supervisor
Project Planning & Scope	R	A	C	I,C
Data Collection & Integration	R	A	C	I,C
Data Preprocessing & Cleaning	A	R	C	I,C
Sentiment Analysis Development	A	R	C	I,C
ML Model Design & Training	R	A	C	I,C
Model Evaluation & Validation	A	R	C	C
Documentation & Reporting	A	C	R	I,C
Presentation & Demonstration	A	I	I	I,C
Ethical & Compliance Checks	I	I	C	I,C
Final Submission & Review	R	A	C	I,C

12. Planning



13. Milestones

▪ FYP-I Evaluation

- **Project Initialization** : Defined problem statement, research objectives, and team roles. Finalized data sources and began initial literature review.
- **Data Collection** : Collected historical OHLCV data for HBL from yfinance and gathered financial news articles from Dawn, Business Recorder, and The News.
- **Data Preprocessing** : Normalized stock data and cleaned news text using tokenization, lemmatization, and stopword removal.
- **Sentiment Annotation** : Applied FinBERT and Loughran-McDonald Dictionary to assign sentiment scores (positive/neutral/negative) to news articles.
- **Feature Engineering** : Merged sentiment scores with stock data by date. Created lag features and structured dataset for model input.
- **Model Design** : Designed LSTM model with attention mechanism. Finalized hyperparameters and evaluation metrics.
- **Model Training** : Trained and tuned the LSTM model using 5-fold cross-validation. Tracked performance via RMSE, MAPE, and directional accuracy.

▪ FYP-II Evaluation :

- **Evaluation** : Assessed model on test data. Conducted error analysis and validated prediction accuracy (target: 85–90%).
- **Research Validation** : Compared results with related studies. Performed statistical validation and finalized findings.
- **Report Writing** : Drafted full FYP-II research report including all sections, visuals, and references.
- **Supervisor Review** : Revised report based on supervisor feedback. Finalized formatting and citations.
- **Final Submission** : Submitted report, prepared slides, and rehearsed presentation for defense.
- Presented findings to the evaluation panel, answered questions, and completed project defense

14. Estimated Budget

- Data access: PKR 15,000 (PSX APIs, news subscriptions)
- Hardware/software: PKR 10,000 (GPU rental, licenses)
- Contingency: PKR 5,000 (unexpected costs)
- Total : PKR 30,000

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